

Practicals\Task2a.cpp

```
1  #include <iostream>
2  #include <vector>
3  using namespace std;
4
5  class Class
6  {
7      private:
8          string name;
9          int rollno;
10         float marks;
11         vector<Class> studentList;
12
13     public:
14         void display();
15         void display(int no_student);
16         void setdata();
17         void setdata(int size);
18 };
19
20 void Class::display()
21 {
22     cout << "Name : " << name;
23     cout << "\nRoll number : " << rollno;
24     cout << "\nMarks : " << marks << endl;
25 }
26
27 void Class::display(int no_student)
28 {
29     for(int i = 0; i < no_student; i++)
30     {
31         studentList[i].display();
32         cout << string(40, '=') << endl;
33     }
34 }
35
36 void Class::setdata(){}
37
38 void Class::setdata(int size)
39 {
40     studentList.resize(size);
41     for (int i = 0; i < size; i++)
42     {
43         cout << "\n--- Student " << i+1 << " ---" << endl ;
44         cout << "Enter name : " ; cin >> studentList[i].name ;
45         cout << "Enter roll number : " ; cin >> studentList[i].rollno ;
46         cout << "Enter marks : " ; cin >> studentList[i].marks ;
47         cout << endl;
48     }
```

```
49  }
50
51  void menu()
52  {
53      cout << "1. Create class & Enter Data " << endl ;
54      cout << "2. Display Data " << endl ;
55      cout << "3. Exit " << endl ;
56  }
57
58  int getChoice()
59  {
60      int choice;
61      cout << "Enter your choice : " ; cin >> choice ;
62      return choice ;
63  }
64
65  int main()
66  {
67      int size = 0;
68      Class AIML_C;
69      string classname;
70
71      while(true)
72      {
73          system("cls");
74          menu();
75          int choice = getChoice();
76
77          if(choice == 1)
78          {
79              cout << "Enter class name : " ; cin >> classname;
80              cout << "Enter class size : " ; cin >> size;
81
82              AIML_C.setdata(size);
83
84              cout << "Enter to continue : " ;
85              cin.ignore(1000, '\n');
86              cin.get();
87          }
88          else if(choice == 2)
89          {
90              if(size == 0)
91              {
92                  cout << "\033[91mError: Class is empty\033[0m\n" ;
93              }
94              else
95              {
96                  cout << "\n--- Students of " << classname << " ---" << endl;
97                  AIML_C.display(size);
98              }
```

```
99
100     cout << "\033[93mEnter to continue : \033[0m";
101     cin.ignore(1000, '\n');
102     cin.get();
103 }
104 else if(choice == 3)
105 {
106     cout << "\033[92mThanks for using programme!\033[0m\n";
107     exit(0);
108 }
109 else
110 {
111     cout << "\n\033[91mInvalid choice !\033[0m\n";
112     cout << "Enter to continue : ";
113     cin.ignore(1000, '\n');
114     cin.get();
115 }
116 }
117 return 0;
118 }
```